

Matias Bronner

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Education

Brown University, *Sc.B. Computer Science and Economics* Sep 2023 – May 2027
GPA: 3.94/4.00 **Coursework:** Deep Learning, Foundations of AI, Machine Learning for Healthcare, Data Structures & Algorithms, Computer Systems, Applied Cryptography, Investments, Corporate Finance, Econometrics

Experience

Micron Technology May 2026 – Aug 2026
AI/ML Engineer Intern, SSD — Python, Claude, MCP, agentic workflows & tools *Boise, ID*

- Built a Claude-powered agentic system in Python that performs automated root-cause analysis of new firmware bugs, improving diagnostic accuracy **15%** over the existing baseline.
- Connected the agent to internal firmware logs, test results, and debugging utilities through MCP tools, and designed multi-step workflows that decompose each bug into hypothesis generation, evidence gathering, and verification.
- Developed an LLM-powered tool that translates natural-language requests into executable queries and firmware telemetry plots, replacing **hours of manual scripting per analysis** with a single prompt.

St. Luke's Health System May 2025 – Aug 2025
Full Stack Intern — Python, Flask, Selenium, Pandas, GPT-4 *Boise, ID*

- Built and deployed a full-stack Flask app parallelizing scraping (Selenium, Google CSE) and GPT-4 description generation across **40,000+ medical products** via background threading.
- Implemented thread-safe state management with lock-guarded DataFrames and a session-driven review queue for supply chain staff to inspect, edit, and export results without data races.

Brown University, Department of Computer Science Sep 2025 – May 2026
Teaching Assistant, Foundations of AI — Python, PyTorch, JavaScript, TypeScript, Next.js

- Supported **100+ students** across reinforcement learning, adversarial search, and Q-learning; built and debugged PyTorch and NumPy assignments prior to release.
- Built and maintained the course website in Next.js, publishing assignments, lecture materials, and resources for the course.

Projects

Spotify Playlist Transformer — *Python, TensorFlow, Set Transformers*

- Built a title-conditioned set transformer with pooling by multi-head attention that recommends five songs to extend a playlist; trained on **994K** Spotify playlists (**37M+** playlist-track pairs) using a custom WARP ranking loss.

Skin Lesion Classifier — *Python, PyTorch, ConvNeXt*

- Fine-tuned ConvNeXt-Base with LP-FT staging and fused it with a tabular MLP to classify 7 lesion categories; reached **Macro-F1 0.679**, beating image-only (0.65) and tabular-only (0.17) baselines.

Go AI Bot — *Python, PyTorch, NumPy*

- Developed a custom Go bot fusing Monte Carlo Tree Search, Iterative Deepening Search, and custom heuristics; **won the class-wide Brown Go AI competition**, outperforming **70 bots**.

Distributed Key-Value Store — *C++, Docker, Concurrency*

- Built a sharded distributed key-value store in C++ with a dynamic Shardmaster that assigns and rebalances key ranges as servers join and leave; ensured consistency under concurrent access with mutexes and readers-writer locks.

Leadership & Activities

FinTech@Brown Jun 2025 – Present
Treasurer & Web Developer — JavaScript, TypeScript, Next.js *Providence, RI*

- Manage finances, establish partnerships with fintech firms across Providence and Boston, and built the club website.

Hispanic Scholarship Fund Jun 2023 – Present
National Hispanic Scholar & Mentor

- Selected as a National Hispanic Scholar from **120,000+** applicants.
- Guided 5 students through college applications, with placements including Harvard, Yale, USC, and Amherst.

Brown Jazz Band, Boise Philharmonic Youth Orchestra, & BHS Jazz Band Sep 2020 – Present
Percussionist & Drummer

- Performed alongside students and professionals; largest audience of **8,000+**.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, HTML/CSS

AI/ML: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LLMs & agentic systems, fine-tuning, Whisper, NLTK

Web & Frameworks: React, Next.js, Flask, Express.js

Tools: Git, Docker, Selenium, REST APIs, Matplotlib, MCP, Claude Code